console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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retail_sales | *Google ...all

Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
--table overview

SELECT * FROM `practical-3-493417.001.retail_sales` LIMIT 1000;

-- Filter all transactions that occurred in the year 2023.

SELECT*
FROM `practical-3-493417.001.retail_sales`
WHERE EXTRACT(YEAR FROM Date) = 2023
LIMIT 1000;
```

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Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Transaction ID | Date | Customer ID | Gender | Age | Product Category | Quantity |
|---|---|---|---|---|---|---|---|
| 1 | 191 | 2023-10-18 | CUST191 | Male | 64 | Beauty | |
| 2 | 204 | 2023-09-28 | CUST204 | Male | 39 | Beauty | |
| 3 | 230 | 2023-04-23 | CUST230 | Male | 54 | Beauty | |
| 4 | 232 | 2023-02-06 | CUST232 | Female | 43 | Beauty | |
| 5 | 309 | 2023-12-23 | CUST309 | Female | 26 | Beauty | |

Results per page: 50 | 1 – 50 of 998

Job history | Show

16° | Search | 22:03 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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retail_sales | *Google ...all

Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.


SELECT *
FROM `practical-3-493417.001.retail_sales`
WHERE `Total Amount` > (
    SELECT AVG(`Total Amount`)
    FROM `practical-3-493417.001.retail_sales`);
```

This script will process 343.32 KB when run.

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Total_Revenue |
|---|---|
| 1 | 456000 |

Results per page: 50 | 1 – 1 of 1

Job history | Show

Search | 22:02 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- Calculate the total revenue (sum of Total Amount).
SELECT
SUM(`Total Amount`) Total_Revenue
FROM `practical-3-493417.001.retail_sales`;
```

This script will process 343.32 KB when run.

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Total_Revenue |
|---|---|
| 1 | 456000 |

Results per page: 50 | 1 – 1 of 1

Job history | Show

Search | 22:01 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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retail_sales | *Google ...all

Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- Display all distinct Product Categories in the dataset.
SELECT
  DISTINCT (`Product Category`)

FROM `practical-3-493417.001.retail_sales`;

-- For each Product Category, calculate the total quantity sold
```

Query completed

Search BigQuery resources

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Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Product Category |
|---|---|
| 1 | Beauty |
| 2 | Clothing |
| 3 | Electronics |

Results per page: 50 | 1 – 3 of 3

Job history | Show

Search | 22:00 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- For each Product Category, calculate the total quantity sold

SELECT
`Product Category`,
SUM(quantity) Total_Quantity
FROM `practical-3-493417.001.retail_sales`
GROUP BY ALL;
```

Query completed

Using on-demand processing quota

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Product Category | Total_Quantity |
|---|---|---|
| 1 | Beauty | 771 |
| 2 | Clothing | 894 |
| 3 | Electronics | 849 |

Results per page: 50 | 1 – 3 of 3

Job history | Show

Search | 21:58 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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retail_sales | *Google ...all

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Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
/* Create a column called Age_Group that classifies customers as:
- 'Youth' (<30)     - 'Adult' (30-59)     - 'Senior' (60+)*/

SELECT
        `Customer ID`,
        Age,
      CASE
          WHEN Age <30 THEN 'Youth'
          WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
          ELSE 'Senior'
      END AS Age_Group
  FROM `practical-3-493417.001.retail_sales`;
```

Using on-demand processing quota

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Customer ID | Age | Age_Group |
|---|---|---|---|
| 1 | CUST191 | 64 | Senior |
| 2 | CUST204 | 39 | Adult |
| 3 | CUST230 | 54 | Adult |

Results per page: 50 | 1 – 50 of 1000

Job history | Show

Search | 21:57 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

Google Cloud | Practical 3 | Search (/) for resources, docs, products and more | Search

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retail_sales | *Google ...all

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Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- For each Gender, count how many high-value transactions occurred (where Total Amount > 500).Expected Output: Gender, High_Value_Transactions

SELECT
        Gender,
        COUNT(`Total Amount`) High_value
FROM `practical-3-493417.001.retail_sales`
WHERE `Total Amount`> 500
Group BY ALL;
```

This script will process 343.32 KB when run.

Using on-demand processing quota

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Gender | High_value |
|---|---|---|
| 1 | Female | 155 |
| 2 | Male | 144 |

Results per page: 50 | 1 – 2 of 2

Job history | Show

Search | 21:45 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- For each Product Category, show only those categories where the total revenue exceeds 5,000.Expected Output: Product_Category, Total_Revenue

SELECT
      `Product Category`,
      SUM(`Total Amount`) Total_Revenue
FROM `practical-3-493417.001.retail_sales`
GROUP BY ALL
HAVING Total_Revenue > 5000;
```

This script will process 343.32 KB when run.

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Product Category | Total_Revenue |
|---|---|---|
| 1 | Beauty | 143515 |
| 2 | Clothing | 155580 |
| 3 | Electronics | 156905 |

Results per page: 50 | 1 – 3 of 3

Job history | Show

Search | 21:41 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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retail_sales | *Google ...all

Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- Display a new column called Unit_Cost_Category that labels a transaction as:

SELECT
        `Transaction ID`,
        `Price per Unit`,

CASE
    WHEN `Price per Unit` < 50 THEN 'Cheap'
    WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
    ELSE 'Expensive'
  END Unit_Cost_Category
FROM `practical-3-493417.001.retail_sales`;
```

Query completed

Search BigQuery resources

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Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Transaction ID | Price per Unit | Unit_Cost_Category |
|---|---|---|---|
| 1 | 191 | 25 | Cheap |
| 2 | 204 | 25 | Cheap |
| 3 | 230 | 25 | Cheap |

Results per page: 50 | 1 – 50 of 1000

Job history | Show

Search | 21:40 2026/04/15

console.cloud.google.com/bigquery?project=practical-3-493417&ws=!1m16!1m4!4m3!1spractical-3-493417!2s001!3sretail_sales!1m10!12m5!1m3!1spractical-3-493417!2safrica-south1!3s9ed6...

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retail_sales | *Google ...all

Google Big_rRetail | Run | Schedule | Open in | More | Save query | Download | Share

```
-- Display all transactions from customers aged 40 or older and add a column Spending_Level

SELECT
        `Customer ID`,
        Age,
        `Total Amount`,

CASE
    WHEN `Total Amount` >1000 THEN 'High'
    ELSE 'Low'
  END  Spending_Level

FROM `practical-3-493417.001.retail_sales`
WHERE Age >=40;
```

This script will process 343.32 KB when run.

Using on-demand processing quota

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

| Row | Customer ID | Age | Total Amount | Spending_Level |
|---|---|---|---|---|
| 46 | CUST834 | 56 | 60 | Low |
| 47 | CUST871 | 62 | 60 | Low |

Results per page: 50 | 1 – 50 of 558

Job history | Show

Search | 21:38 2026/04/15